Workshop 2 – Day 2

The Merged Model Framework

Disaggregation



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Plan for today

1. **Get the model running**
   Follow the setup guide and submit the form once you have opened the scenario report.

2. **Finish the Day 1 exercises**
   Run the public investment and export growth scenarios and interpret the results.

3. **Walk through a disaggregation**
   Split agricultural GDP into rice and non-rice agriculture.

4. **Try the same method yourself**
   Split agricultural exports into rice and non-rice exports and check that the split reproduces the original totals.

Get the model running

Open the setup guide

Find the **Merged Model setup guide (PDF)** in the workshop materials.

Tell us when you are ready

Once you have opened the scenario report, enter your name in the form and submit it once.

Mark the guide as complete



Scan to open the form

Day 1 Exercises

Exercise 1: Change public investment

Create a scenario with infrastructure investment growth of 10% in 2027.

1. Keep the baseline unchanged. In `shocks/scenario.jl`, set infrastructure investment growth to `0.100` for 2027.
2. Save your changes, run the model with scenarios enabled, and open the comparison report.
3. Compare public investment (IVG), private investment (IVP), and government borrowing (BRG) with the baseline.
4. Write down one difference and use a model equation to explain it.

When you finish

Change the size of the investment increase and compare the results again.

Exercise 2: Change export growth

Create a scenario with industrial export growth of -3% in 2027.

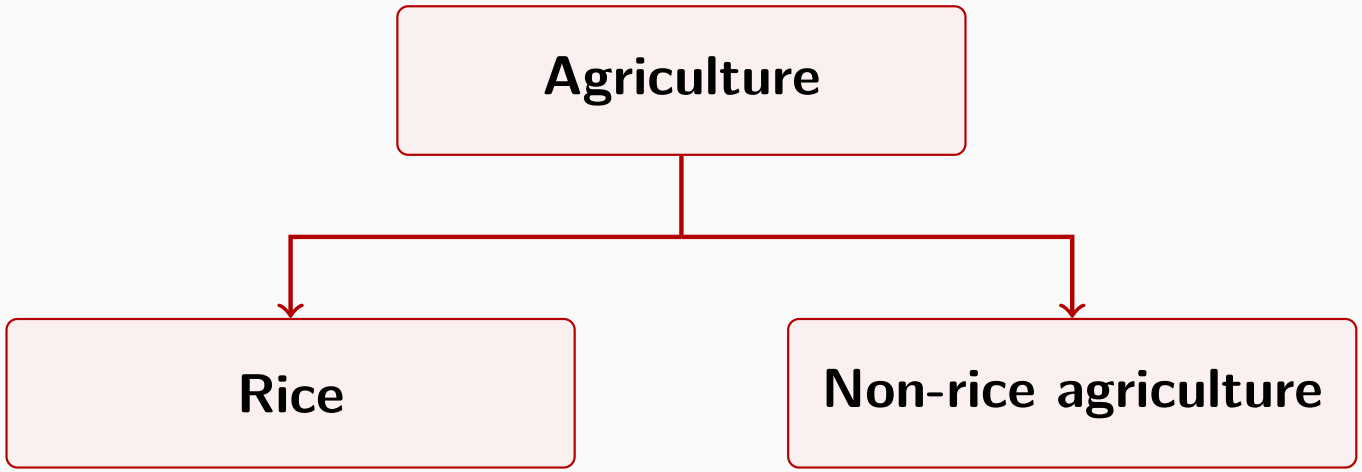
1. Start from the baseline assumptions. The only override in this scenario should be industrial export growth of `-0.030` in 2027.
2. Before running, write down which results you expect to change.
3. Run the scenario and compare total exports (X), private domestic credit (DCP), and private foreign debt ($NFDP$) with the baseline.
4. Use a model equation to explain one result. Did it match your prediction?

When you finish

Extend the lower export growth to another year and compare the results again.

Disaggregation Walkthrough

Splitting agricultural GDP



The two groups add up to agricultural GDP.

Check the split in two runs

Run 1: Reproduce the original path

Give both components the original agricultural growth rate in every year. Does agricultural GDP match the original run throughout the projection?

Run 2: Change rice growth

Lower rice value-added growth in 2027. Before running, predict how much agricultural GDP growth will change. What determines the size of the effect?

Disaggregation Exercise

Split agricultural exports

Repeat the GDP walkthrough for exports. Split agricultural exports into **rice exports** and **non-rice agricultural exports**.

1. In `data/assumptions.jl`, assign 30% of base-year agricultural exports to rice and 70% to non-rice agriculture. Use these illustrative shares and give both groups the original agricultural export growth rate.

2. In `data/projection_paths.jl`, project the two groups separately. Add them back together and use the growth rate of their sum for agricultural exports.

3. Run the model. Check that both agricultural exports and total exports match the original run in every year.

Once your paths match

Lower rice export growth in 2027. Before you rerun, predict what will happen to agricultural exports and total exports.

Thank you for today